TRACK: SOFTWARE WEB DEVELOPMENT

Movie App

Graduation Project Documentation

TEAM MEMBERS:

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Project Idea

The **Movie App** (titled on-screen as *Movie Explorer*) is a client-side single-page web application designed to give users an effortless and interactive platform for media discovery. It serves as a unified digital dashboard where movie enthusiasts can explore cinematic metadata seamlessly in real-time. By utilizing a modern frontend architecture, the application aims to reduce browsing friction, providing an immersive data display that updates fluidly as users interact with the interface.

The core focus of this application is accessibility and performance. It consolidates scattered movie parameters into an organized visual grid system. Users can browse the most popular releases worldwide, execute specific search queries to find exact movie information, view deep metadata parameters (such as release dates, summaries, and rating calculations), and build custom personal collections through an instant local client storage state saving mechanism.

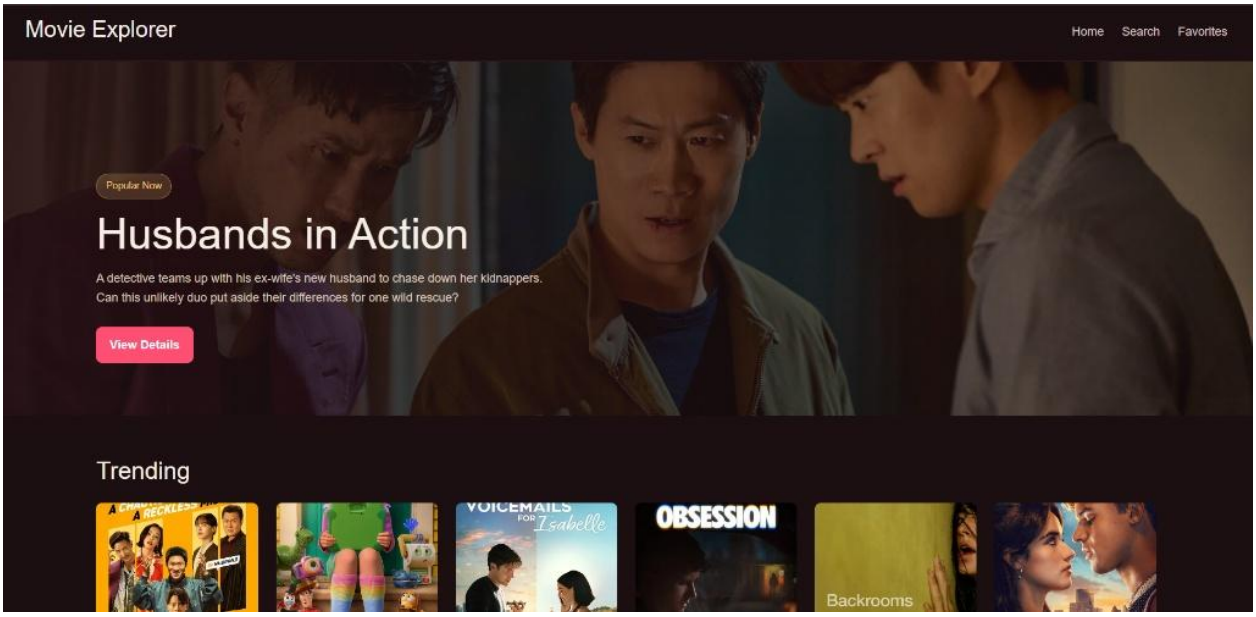
Technologies Used

To ensure high performance, clean component modularity, and smooth single-page user interactions, the platform was developed using the following technological stack:

- **React.js**
- **React Router DOM**
- **Bootstrap**
- **Axios**
- **TMDB API (The Movie Database)**
- **HTML5**
- **CSS3**
- **JavaScript (ES6+)**

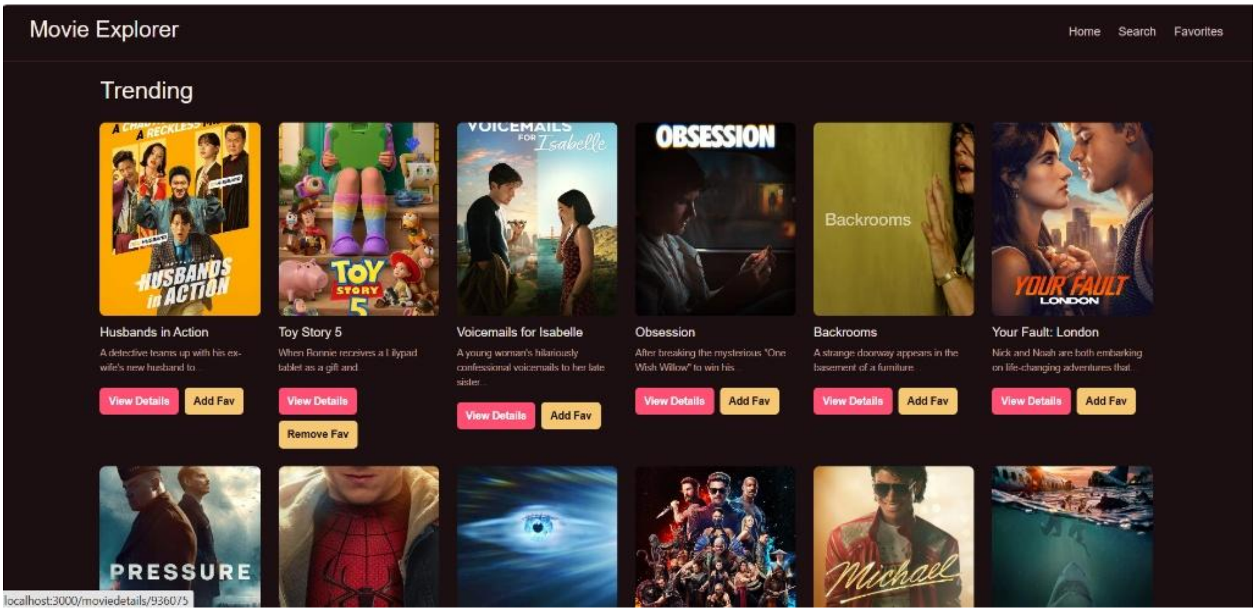
Project Screenshots

Home Page View



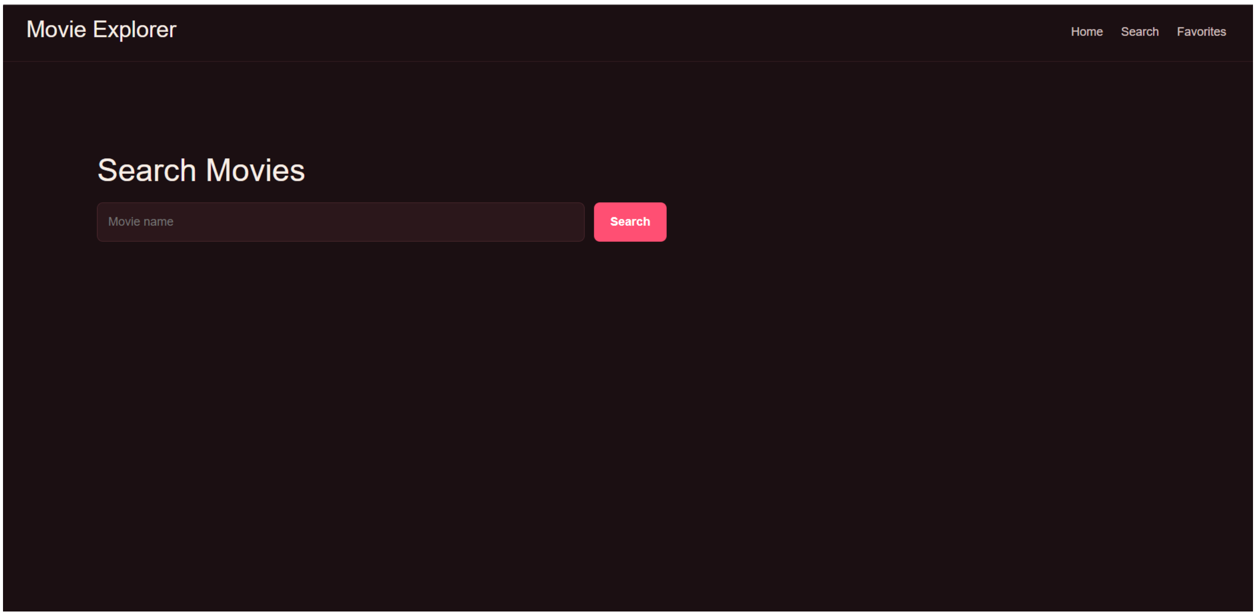
Description: The application landing dashboard featuring a premium dark interface, a high-impact responsive hero banner highlighting the featured trending title ("Husbands in Action"), and a direct view callout action link button.

Trending Movies View



Project Screenshots (Continued)

Search Movies Page



Description: A minimal search console view providing an isolated layout with a text input box ("Movie name") that reads inputs via states to fire dynamic conditional search operations.

Movie Details Page

Description: An in-depth single item detail interface displaying detailed poster artwork columns alongside specific response criteria including localized release dates and explicit remote audience rating attributes.

Challenges and Future Improvements

Challenges Encountered

- **Team Coordination and Communication:** Managing independent feature branches and pull request logs across four developers required setting clean directory layouts to prevent code merge conflicts.
- **API Integration:** Asynchronously acquiring nested JSON payload tracks from external servers required establishing rigorous promise controls to map data strings cleanly into structural view components.
- **Routing Management:** Restructuring navigation parameters within a single-page flow to pass individual numeric target tags flawlessly between discovery indices and item detail panels.
- **Responsive Design:** Adapting the customized cinematic media matrices to scale seamlessly from multi-column wide viewports down to tight single-stack responsive layouts.

Future Improvements Roadmap

- **User Login System:** Migrating local context data into remote databases via server-side session authentication tokens (JWT) to preserve personalized user details across devices.
- **Online Ticket Booking System:** Scaling component properties to fetch operational theater schedules and secure local reservation metrics natively within the application.
- **Watchlist Feature:** Adding customizable folder sorting elements to allow secondary bookmark trackers distinctly separated from favorites tables.
- **Dark Mode toggle:** Introducing a global user interface color context switcher switch allowing easy dynamic standard palette inversions.

Conclusion

The **Movie App** graduation project successfully delivers a completely functional single-page media discovery and organization platform. The development team successfully configured a responsive component hierarchy using React.js that interfaces reliably with live entertainment endpoints via Axios.

All core criteria outlined in the system requirements—including interactive trending dashboard interfaces, smooth parameters single-page routing flows, state-bound search consoles, and automated browser localized storage operations—were integrated without errors. The application architecture establishes a robust, highly extensible framework, making it completely prepared for deployment and future service scaling.